

Evidence Report

OPNsense Firewall Policy Testing, Logging and Packet Analysis

Item	Detail
Student name	RISA THOMAS KOILA
Student ID	C11/26/FCDF/17156
Course	WADF105 Network Security Fundamentals, ICDFa
Date completed	6 October 2026
Virtual machines	Ubuntu client (icdfa-nslab-client-v1) and OPNsense 26.7 firewall (icdfa-nslab-firewall-v1)
Platform	Oracle VirtualBox; firewall WAN on NAT, LAN and client on Internal Network ICDFa-LAN OPNsense LAN 10.10.10.1/24

Purpose: demonstrate a working baseline, apply two narrow firewall blocks, correlate firewall logs with packet captures and states, then restore normal connectivity.

1. Lab Overview

Objective. The laboratory tests top-down OPNsense rule processing by blocking only IPv4 ICMP to 1.1.1.1 and only outbound TCP port 80, while keeping DNS and HTTPS available. The guide specifies the OPNsense LAN as 10.10.10.1/24 and the Ubuntu client as a 10.10.10.x host.

Key test logic. A specific block must be above the broad Default allow LAN to any rule; otherwise, the broad allow can match first and the narrow block will not take effect.

Item	Observed / Required Value
OPNsense LAN	10.10.10.1/24
Ubuntu client IPv4	10.10.10.237
Default route	via 10.10.10.1, interface enp0s3
Temporary block 1	LAB2 BLOCK ICMP TO 1.1.1.1
Temporary block 2	LAB2 BLOCK OUTBOUND HTTP

Item	Observed / Required Value
Automatic outbound NAT	Enabled / automatic source NAT rule generation

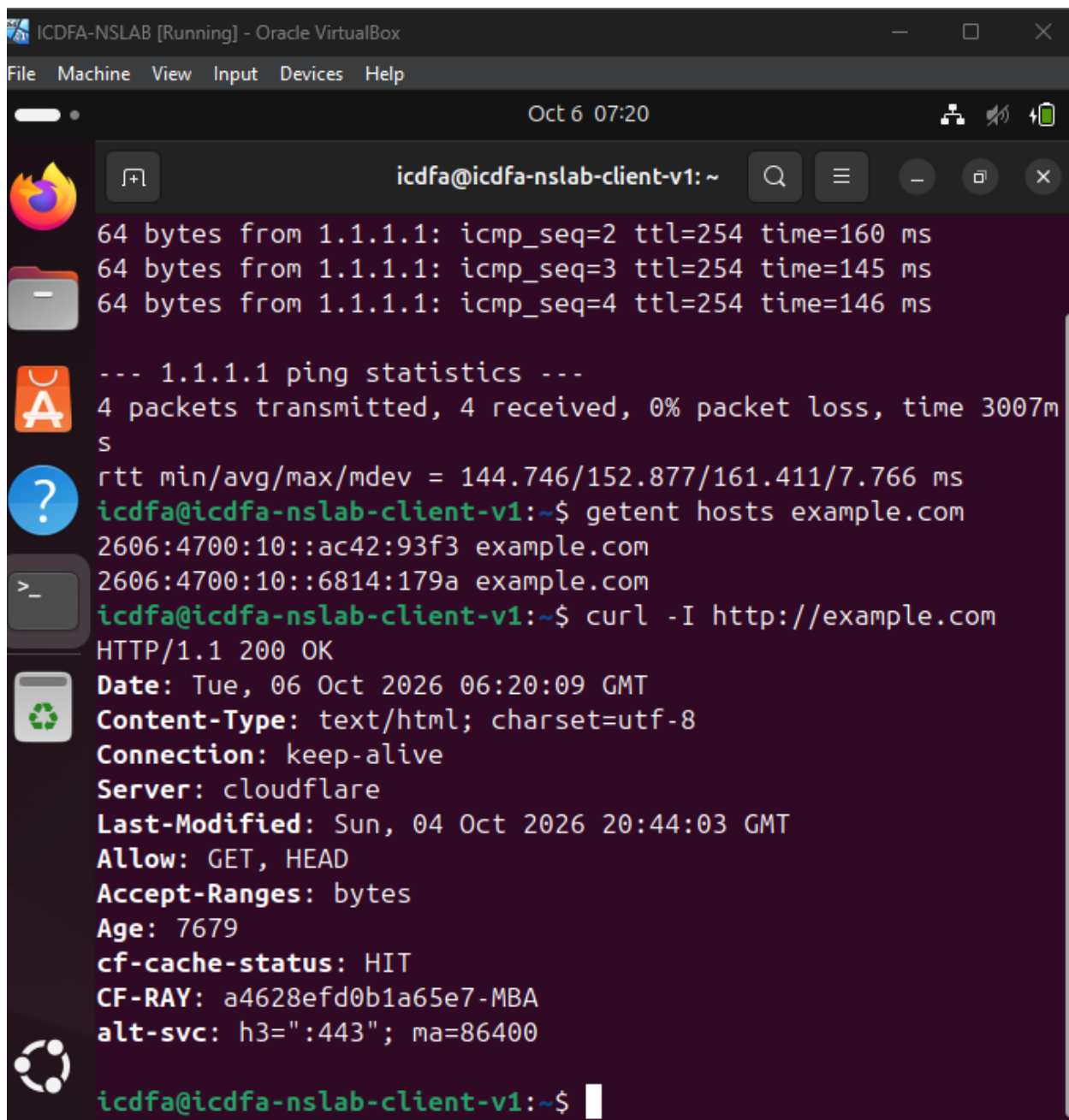
2. Evidence Checklist

Evidence	Requirement	Supplied proof	Status
E1	Baseline ping, DNS, HTTP and HTTPS	Demonstrated from supplied command output	Complete
E2	Both LAB2 block rules above broad allow	Two rule-page captures supplied; a single combined page is not visible	Partial
E3	Failed ICMP with DNS/HTTPS success	Single terminal capture	Complete
E4	Failed HTTP with HTTPS success	Single terminal capture	Complete
E5	Live View blocked ICMP and TCP/80	Live View capture with rule-detail modal	Complete
E6	Wireshark blocked traffic and permitted HTTPS	ICMP, HTTP/80 and HTTPS/443 captures	Complete
E7	State table for permitted HTTPS	States filtered to 10.10.10.237	Complete
E8	Rules disabled + connectivity restored	Restored connectivity is shown; final disabled-rule page is	Partial

Evidence	Requirement	Supplied proof	Status
		not visible in supplied set	

E1. Successful baseline ping, DNS, HTTP and HTTPS tests

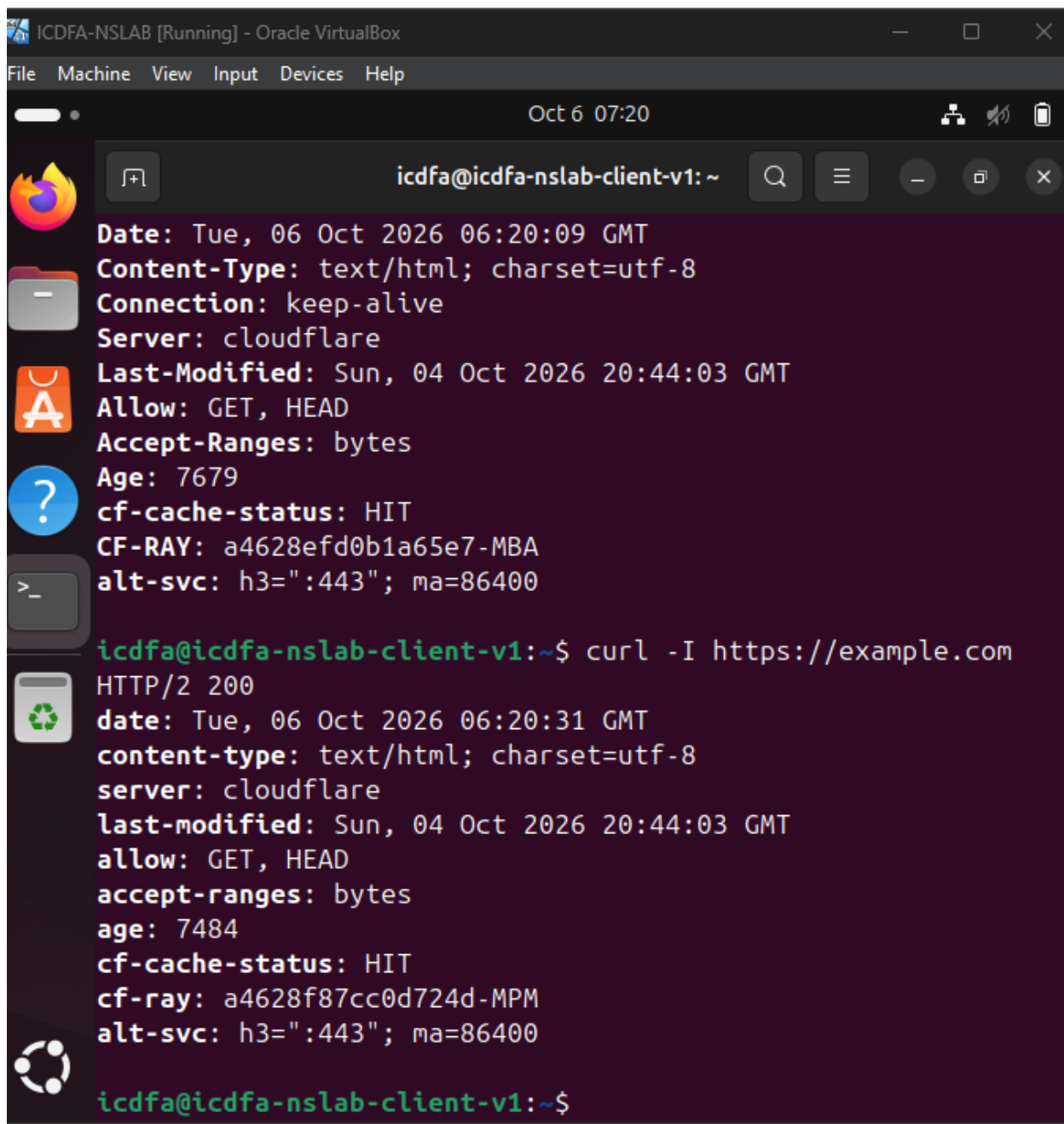
What this proves: The client can reach 1.1.1.1, resolve example.com, retrieve HTTP headers, and retrieve HTTPS headers before/after the controlled policy test. The supplied captures show 0% packet loss, successful DNS resolution, HTTP/1.1 200 OK, and HTTPS HTTP/2 200.



The screenshot shows a terminal window titled "icdfa@icdfa-nsllab-client-v1: ~" within an Oracle VM VirtualBox environment. The terminal displays the output of several network-related commands. It starts with three ping commands to 1.1.1.1, showing successful responses with varying times. This is followed by a "ping statistics" command showing 4 packets transmitted and received with 0% loss. Then, the "getent hosts example.com" command is used to resolve the IP addresses for example.com. Finally, the "curl -I http://example.com" command is executed, displaying the full HTTP response headers, including status 200 OK and various server information.

```
icdfa@icdfa-nsllab-client-v1: ~  
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=160 ms  
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=145 ms  
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=146 ms  
  
--- 1.1.1.1 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 144.746/152.877/161.411/7.766 ms  
icdfa@icdfa-nsllab-client-v1:~$ getent hosts example.com  
2606:4700:10::ac42:93f3 example.com  
2606:4700:10::6814:179a example.com  
icdfa@icdfa-nsllab-client-v1:~$ curl -I http://example.com  
HTTP/1.1 200 OK  
Date: Tue, 06 Oct 2026 06:20:09 GMT  
Content-Type: text/html; charset=utf-8  
Connection: keep-alive  
Server: cloudflare  
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT  
Allow: GET, HEAD  
Accept-Ranges: bytes  
Age: 7679  
cf-cache-status: HIT  
CF-RAY: a4628efd0b1a65e7-MBA  
alt-svc: h3=":443"; ma=86400  
icdfa@icdfa-nsllab-client-v1:~$
```

Figure E1-1. The Ubuntu client shows successful ping to 1.1.1.1, DNS resolution for example.com, and an HTTP 200 response.



```
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 6 07:20
icdfa@icdfa-nslab-client-v1: ~
Date: Tue, 06 Oct 2026 06:20:09 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 7679
cf-cache-status: HIT
CF-RAY: a4628efd0b1a65e7-MBA
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 06:20:31 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 7484
cf-cache-status: HIT
cf-ray: a4628f87cc0d724d-MPM
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$
```

Figure E1-2. The Ubuntu client shows a successful HTTPS request to example.com with an HTTP/2 200 response.

E2. LAN rules showing the two narrow LAB2 block rules

What this proves: The supplied OPNsense rule captures show the two intended narrow rule definitions: one for ICMP to 1.1.1.1/32 and one for TCP destination port HTTP (80). The guide requires both to sit above Default allow LAN to any.

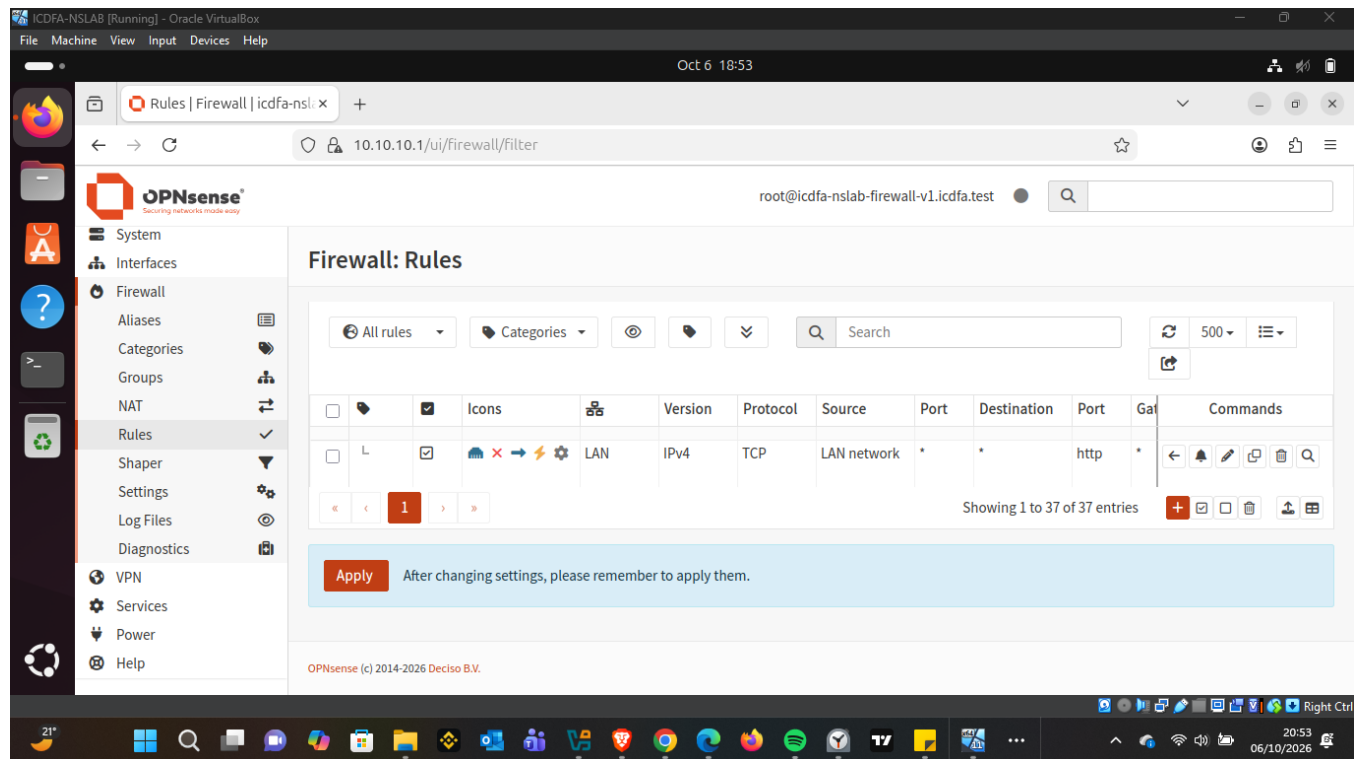


Figure E2-1. The OPNsense rules page visibly shows the TCP/HTTP rule targeting destination port http.

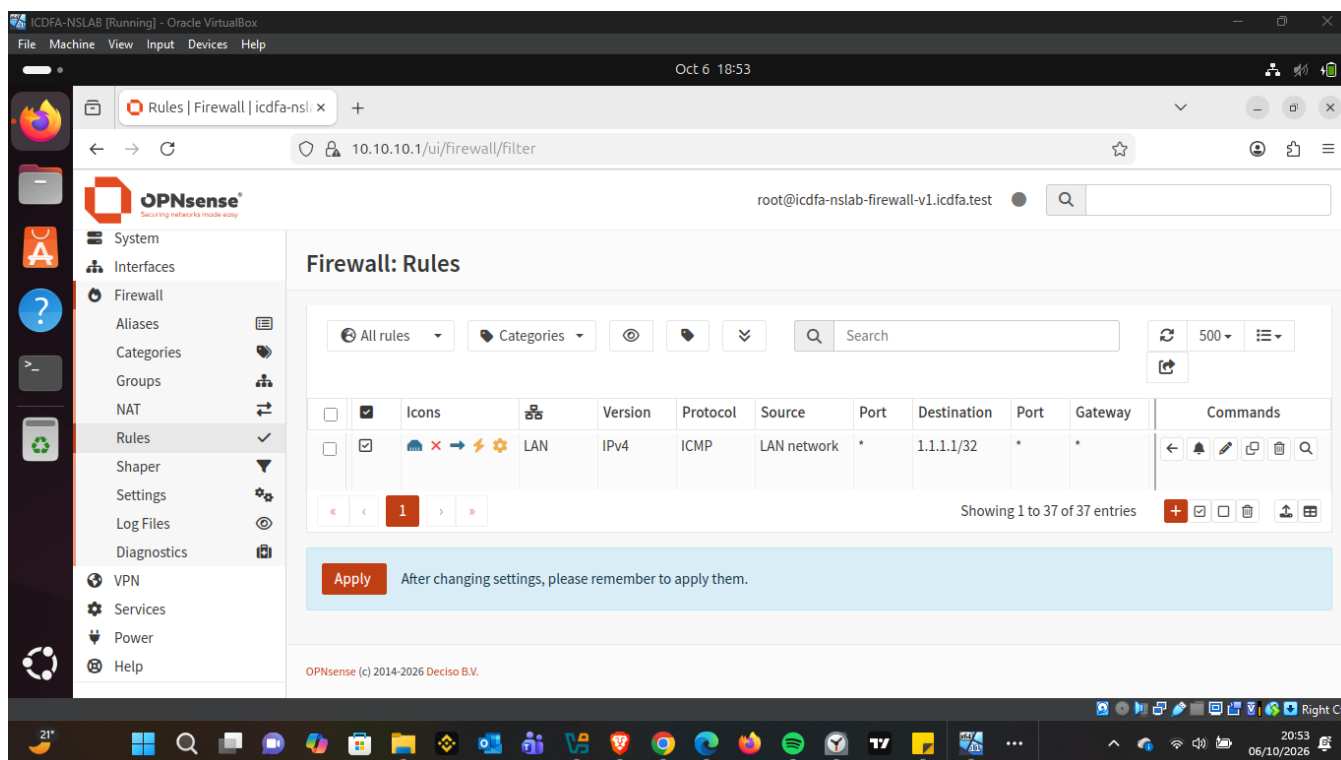
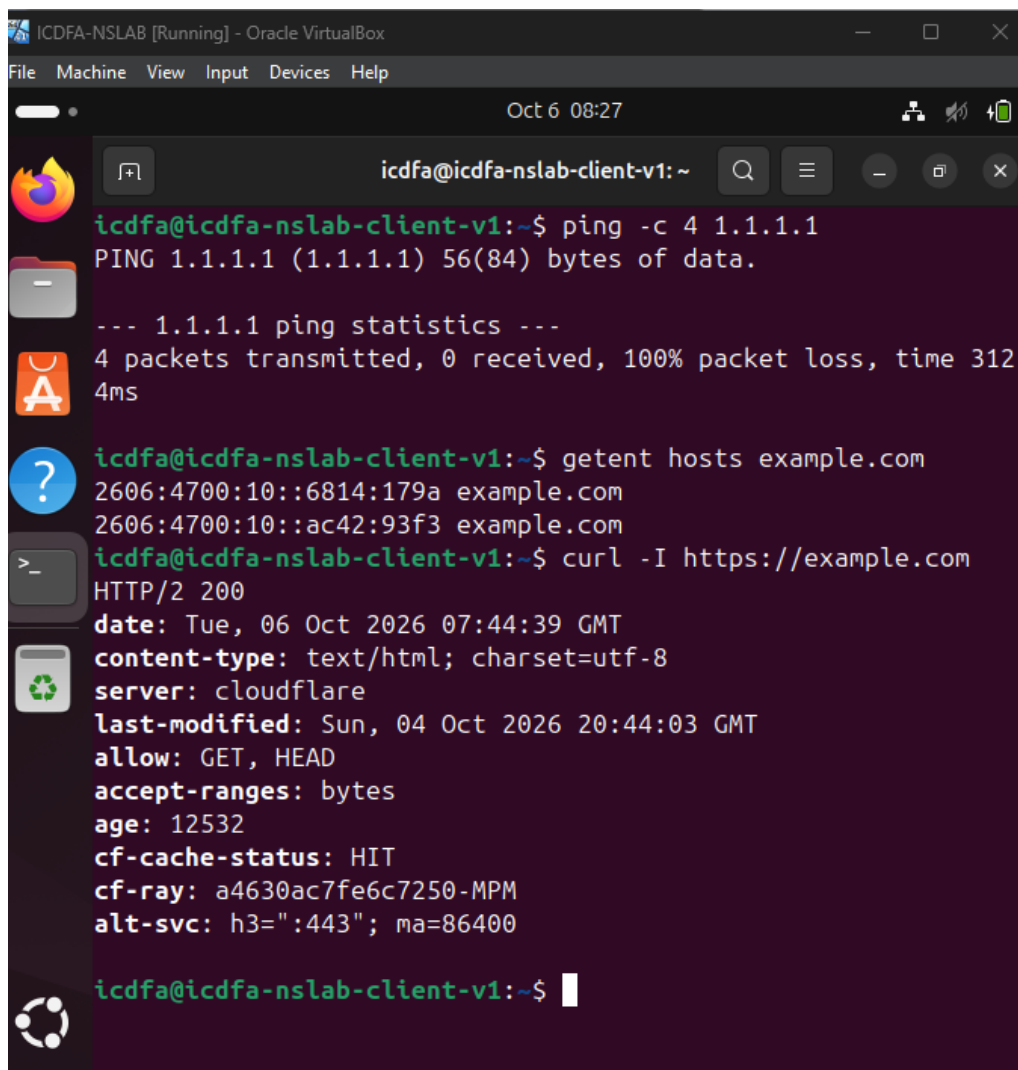


Figure E2-2. The OPNsense rules page visibly shows the IPv4 ICMP rule targeting 1.1.1.1/32.

E3. Failed ICMP test with successful DNS or HTTPS evidence

What this proves: The ping to 1.1.1.1 records 100% packet loss, while `getent hosts example.com` resolves the hostname and `curl -I https://example.com` returns HTTP/2 200, demonstrating protocol-specific filtering rather than total loss of internet connectivity.



```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3124ms

icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com

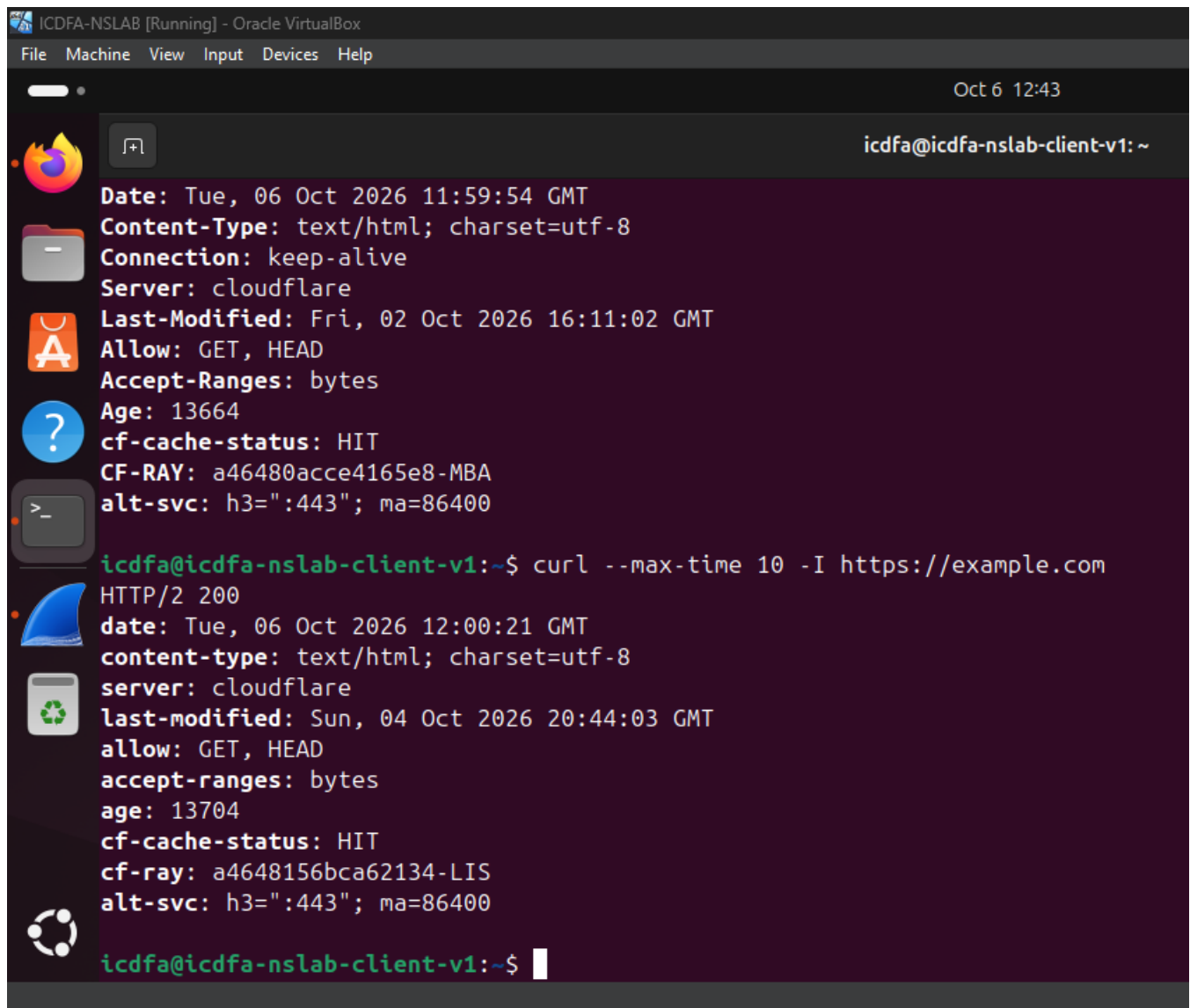
icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 07:44:39 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 12532
cf-cache-status: HIT
cf-ray: a4630ac7fe6c7250-MPM
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$
```

Figure E3-1. ICMP to 1.1.1.1 is fully blocked while DNS resolution and HTTPS remain successful.

E4. Failed HTTP test and successful HTTPS test

What this proves: The HTTP request to port 80 times out while the HTTPS request to port 443 returns response headers, showing that the second rule blocks only unencrypted web traffic.



The screenshot shows a terminal window titled "ICDFA-NSLAB [Running] - Oracle VirtualBox". The terminal output displays the results of two web requests. The first request is an HTTP GET to port 80, which times out. The second request is an HTTPS GET to port 443, which returns a successful response with various headers.

```
icdfa@icdfa-ns-lab-client-v1: ~
Date: Tue, 06 Oct 2026 11:59:54 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Fri, 02 Oct 2026 16:11:02 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 13664
cf-cache-status: HIT
CF-RAY: a46480acce4165e8-MBA
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 12:00:21 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13704
cf-cache-status: HIT
cf-ray: a4648156bca62134-LIS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$
```

Figure E4-1. The HTTP request times out while HTTPS returns an HTTP/2 200 response.

E5. OPNsense Live View entries for blocked ICMP and TCP port 80 traffic

What this proves: The OPNsense Live View capture is filtered to LAB2 traffic and visibly shows red block entries for both LAB2 BLOCK OUTBOUND HTTP and LAB2 BLOCK ICMP TO 1.1.1.1; the detail modal confirms action [block], destination 1.1.1.1, direction in, and interface em1 for the selected event.

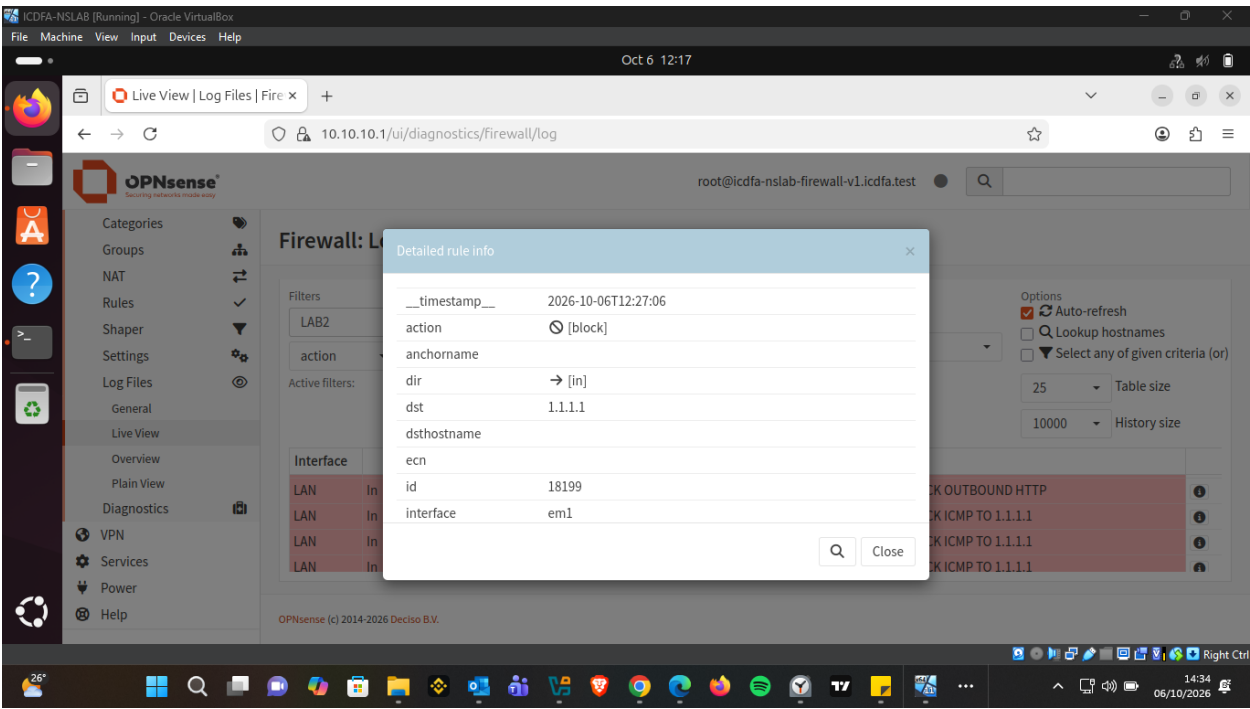


Figure E5-1. OPNsense Live View shows blocked LAB2 ICMP and HTTP traffic, with the selected event identified as a block to 1.1.1.1.

E6. Wireshark evidence for blocked traffic and permitted HTTPS

What this proves: Wireshark provides the packet-level view that matches the firewall behavior: ICMP echo requests are visible without replies, TCP/80 SYN packets are retransmitted, and TCP/443 shows an established encrypted TLS exchange.

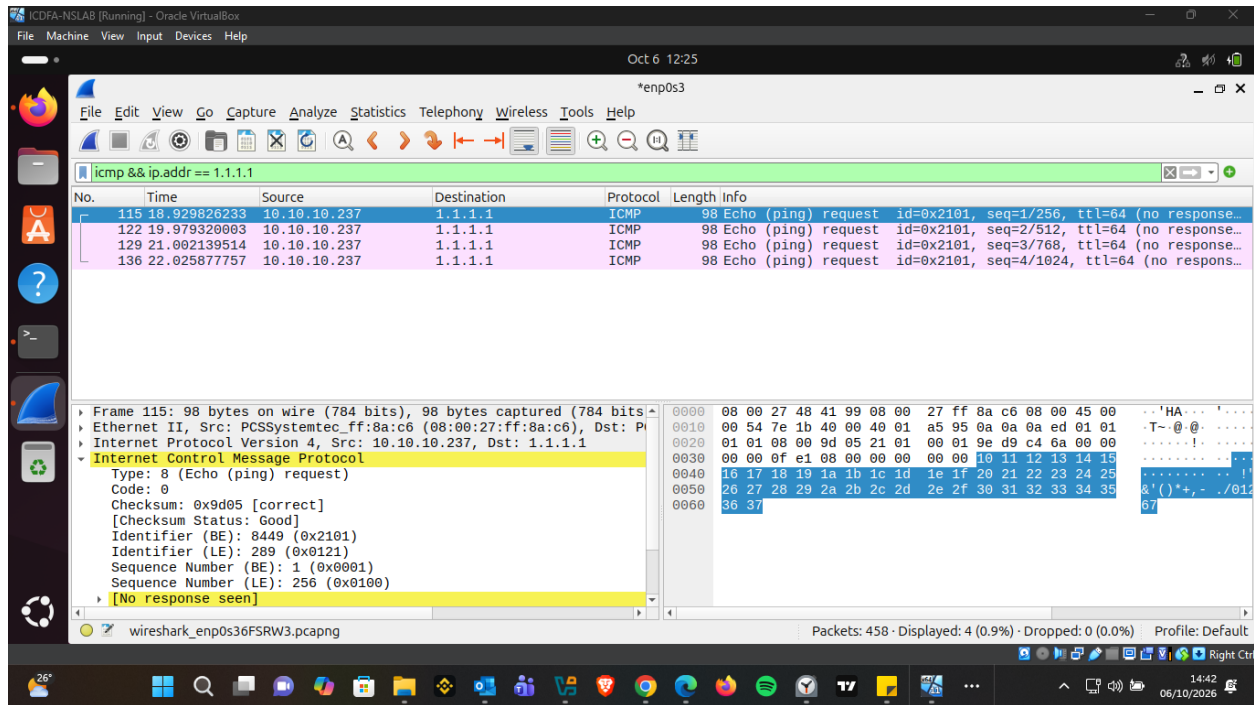


Figure E6-1. Wireshark shows ICMP echo requests from 10.10.10.237 to 1.1.1.1 with no response seen.

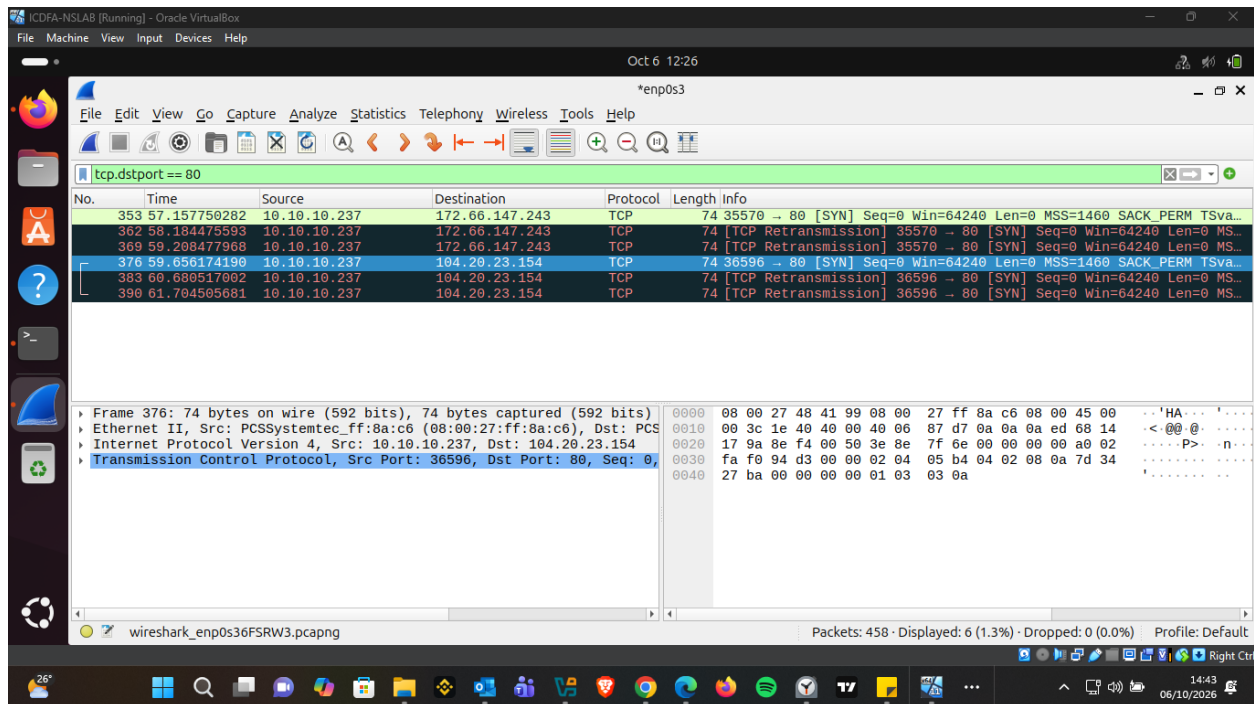


Figure E6-2. Wireshark shows repeated TCP SYN retransmissions to destination port 80.

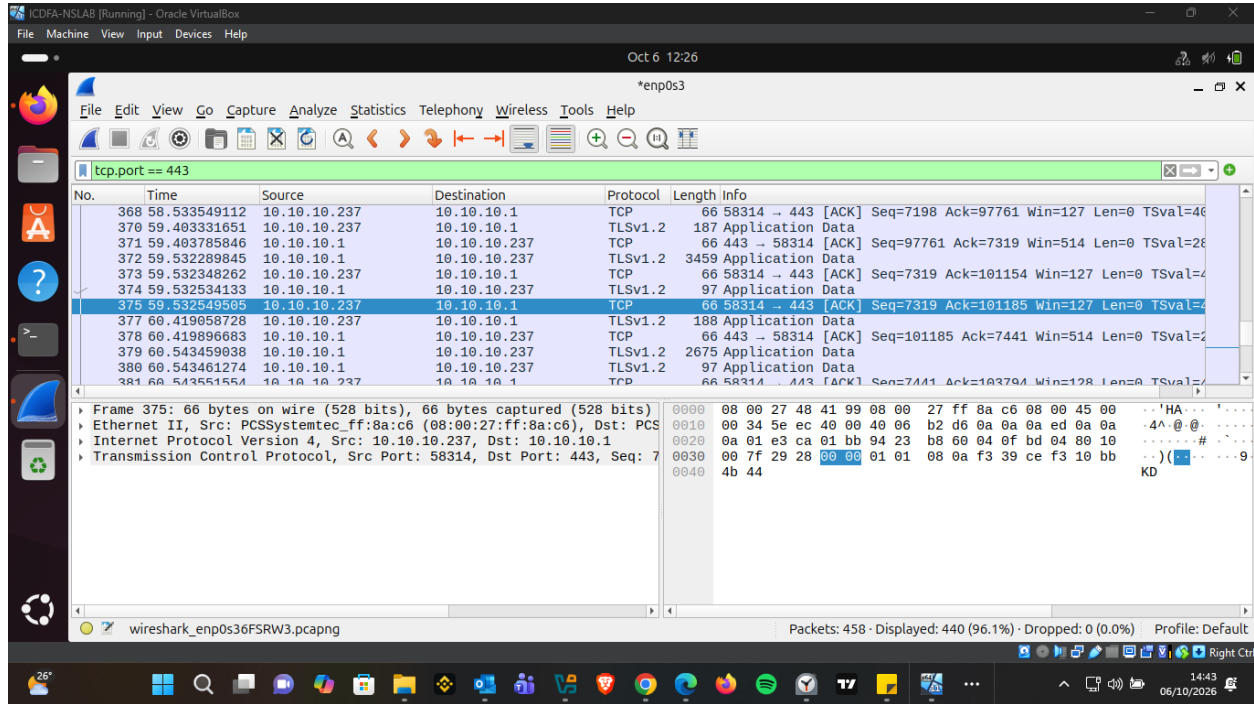
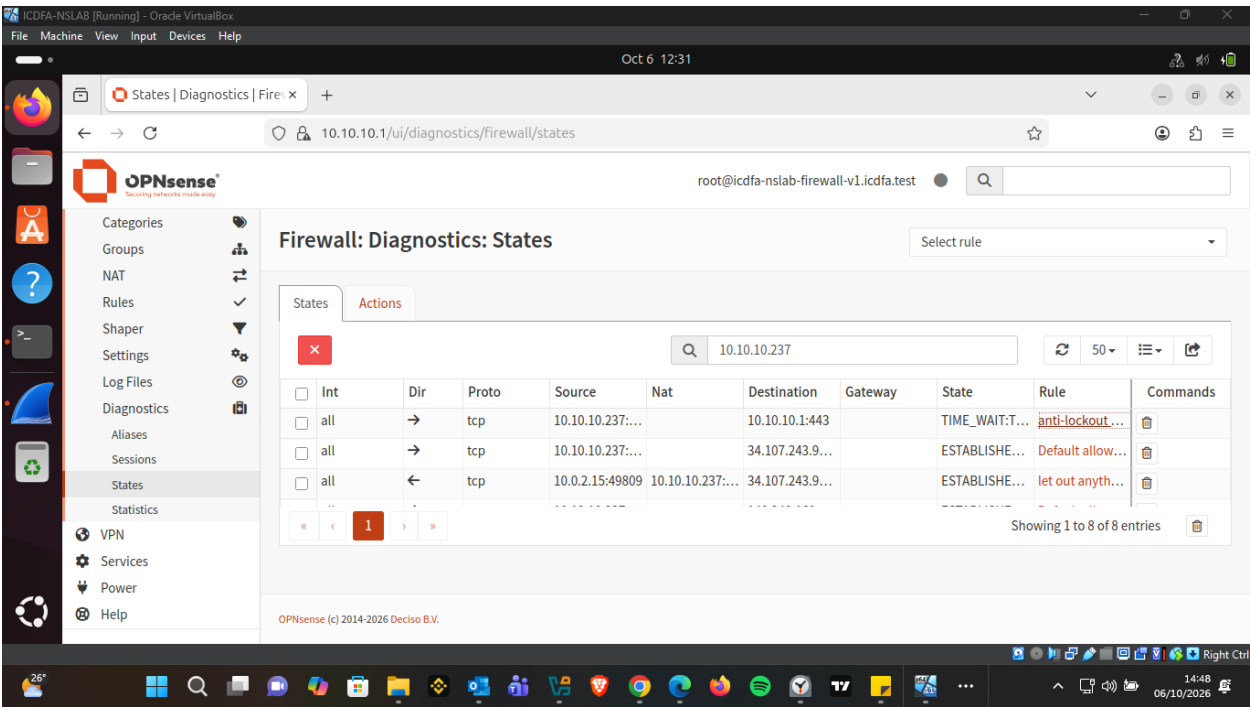


Figure E6-3. Wireshark shows TCP/443 traffic completing the connection and carrying TLSv1.2 application data.

E7. State-table evidence for a permitted HTTPS connection

What this proves: The OPNsense Diagnostics > States page is filtered to 10.10.10.237 and shows established TCP states, including a connection to an external address under the Default allow rule; the state table also exposes the NAT column for the translated flow.



OPNsense Firewall: Diagnostics: States

Search: 10.10.10.237

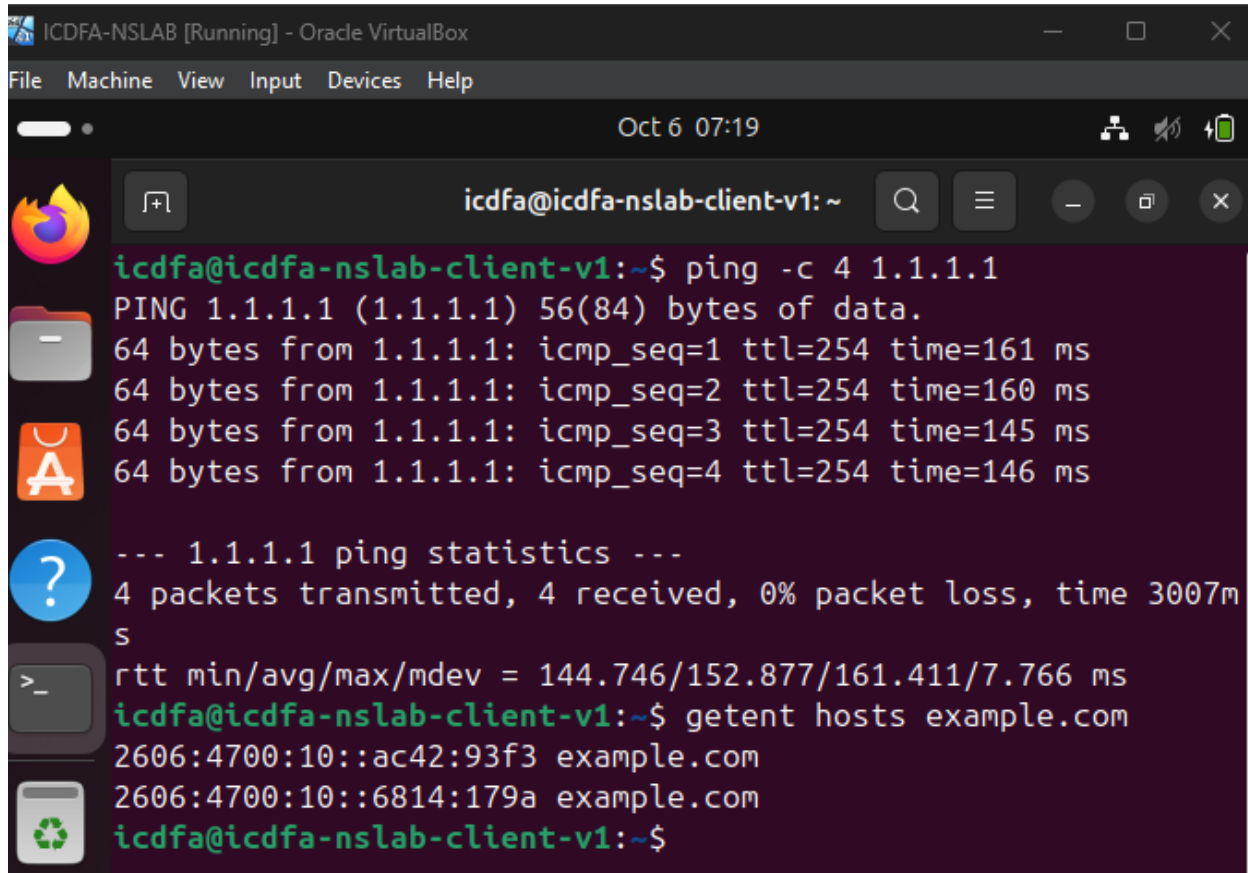
	Int	Dir	Proto	Source	Nat	Destination	Gateway	State	Rule	Commands
☐	all	→	tcp	10.10.10.237:...		10.10.10.1:443		TIME_WAIT:T...	anti-lockout...	
☐	all	→	tcp	10.10.10.237:...		34.107.243.9...		ESTABLISHE...	Default allow...	
☐	all	←	tcp	10.0.2.15:49809	10.10.10.237:...	34.107.243.9...		ESTABLISHE...	let out anyth...	

Showing 1 to 8 of 8 entries

Figure E7-1. OPNsense States shows an established TCP connection from 10.10.10.237 under the default allow path, with NAT information present for the flow.

E8. Final disabled test rules and restored connectivity

What this proves: The supplied terminal captures show connectivity restored after testing, including 0% packet loss to 1.1.1.1 and successful DNS resolution; the screenshot set does not contain a clear final LAN rules page where both LAB2 rules are visibly disabled.



The screenshot shows a terminal window titled "ICDFA-NSLAB [Running] - Oracle VirtualBox". The terminal output is as follows:

```
icdfa@icdfa-nsrab-client-v1: ~  
icdfa@icdfa-nsrab-client-v1:~$ ping -c 4 1.1.1.1  
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.  
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=161 ms  
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=160 ms  
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=145 ms  
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=146 ms  
  
--- 1.1.1.1 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 144.746/152.877/161.411/7.766 ms  
icdfa@icdfa-nsrab-client-v1:~$ getent hosts example.com  
2606:4700:10::ac42:93f3 example.com  
2606:4700:10::6814:179a example.com  
icdfa@icdfa-nsrab-client-v1:~$
```

Figure E8-1. After the test, ping succeeds with 0% packet loss and DNS resolution for example.com works again.

The screenshot shows a terminal window titled "icdfa@icdfa-nslab-client-v1: ~" within an Oracle VM VirtualBox environment. The terminal displays the output of a ping command: `ping -c 4 1.1.1.1`. The output shows four successful pings to 1.1.1.1, each with a TTL of 254 and varying response times between 145ms and 161ms. The ping statistics summary indicates 4 packets transmitted, 4 received, and 0% packet loss. The terminal window is part of a desktop environment with icons for Firefox, a file manager, and other applications visible on the left. The top of the window shows the date and time as "Oct 6 07:19".

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=161 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=160 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=145 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=146 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 144.746/152.877/161.411/7.766 ms
icdfa@icdfa-nslab-client-v1:~$
```

Figure E8-2. A separate restored-state capture again shows successful 1.1.1.1 connectivity with 0% packet loss.

3. Analysis Questions

1. Why must the specific block rules be placed above the broad allow rule?

OPNsense processes interface rules sequentially from top to bottom on a first-match-wins principle when the "Quick" flag is enabled (default behavior). When a packet arrives, the firewall evaluates it against each rule in order until a matching rule is found, at which point the specified action (Pass or Block) is executed immediately and evaluation terminates. If a broad allow rule (e.g., Default allows LAN to any) is placed above a specific block rule, all outgoing traffic will match the top allow rule first. Consequently, the lower block rule will never be evaluated, rendering it completely ineffective.

2. Which five packet attributes are most useful when explaining a firewall decision?

The most useful five are source IP address, destination IP address, protocol, destination port, and traffic direction/interface. Together they describe who is sending, where the traffic is going, what transport is used, which service is targeted, and where the rule is applied. The five primary tuple attributes evaluated by stateful firewalls are:

Source IP Address: Identifies the originating node or network segment (e.g., 10.10.10.237 / LAN net).

Destination IP Address: Identifies the targeted remote host or subnet (e.g., 1.1.1.1).

Protocol: Specifies the Transport Layer or Network Layer protocol (e.g., TCP, UDP, ICMP).

Destination Port: Indicates the specific target service on TCP or UDP (e.g., port 80 for HTTP, port 443 for HTTPS).

Interface and Direction: Specifies where the packet entered or exited the firewall and its vector (e.g., LAN interface, In direction).

3. Why did blocking ICMP not block HTTPS?

ICMP (Internet Control Message Protocol) operates at Layer 3 (Network Layer) of the OSI model and is used primarily for diagnostic utilities such as ping and traceroute. In contrast, HTTPS operates at Layer 7 (Application Layer) encapsulated within TLS/SSL over TCP (Layer 4) on destination port 443. Because OPNsense performs fine-grained protocol-specific filtering, a rule targeting Layer 3 ICMP traffic only drops packets matching IP Protocol 1 (ICMP). It has no effect on Layer 4 TCP transport streams, allowing HTTPS traffic on port 443 to pass through unimpeded.

4. What difference did you observe between the blocked TCP port 80 traffic and permitted TCP port 443 traffic?

Blocked TCP Port 80 Traffic: When curl initiated a connection to port 80, the client sent a TCP [SYN] packet. OPNsense silently dropped this packet according to the Block rule. Because no response ([SYN, ACK] or [RST]) was returned, Wireshark captured multiple TCP Retransmission attempts from the client until the request hit the timeout limit (curl error 28). No entry was created in the OPNsense state table.

Permitted TCP Port 443 Traffic: Traffic sent to port 443 matched the default allow rule. The firewall permitted the [SYN] packet, established a state entry in the state table, and allowed the 3-way TCP handshake to complete successfully. Wireshark captured the handshake followed by encrypted TLS application data exchange, and curl received a valid HTTP/2 200 OK response.

5. What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

icdfa-nslab-client-v1 uses a private, non-routable IPv4 address (10.10.10.237/24 under RFC 1918). Routers on the public Internet drop packets containing private source or destination addresses. Outbound Source NAT (SNAT) on OPNsense solves this by dynamically modifying the IPv4 header of outbound packets leaving the WAN interface. The firewall replaces the client's private source IP (10.10.10.237) with its own WAN interface IP address. It tracks this translation in its NAT table so that when response packets arrive on the WAN interface, OPNsense rewrites the destination IP back to 10.10.10.237 and forwards the packet to the internal client.

6. Why is restoring the original state an important part of a controlled security laboratory?

Restoring the firewall and client configuration to a known-good baseline is an essential practice in security engineering for several reasons:

Elimination of Residual Effects: Stale test rules or active state connections can interfere with subsequent tests, causing unexplained failures or false positives.

Environment Predictability: Ensures that subsequent security labs start from a standardized baseline state.

Production Parity: In operational environments, leaving temporary test or debug rules active introduces security vulnerabilities or unintended outages.

4. Completion Assessment

Demonstrated in the supplied evidence: working connectivity, a protocol-specific ICMP block, a port-specific HTTP block, firewall log correlation, packet-level evidence, a permitted HTTPS state, and restored connectivity.